POLICY

APPS

ENGINE

CORE

LOGGER

ASSERT

TIMESYSTEM

HIRESCLOCK

IALLOCATOR (INTERFACE)

DEBUGALLOC

LINEARARENA

SUBSYSTEMREGISTRY

PLATFORM/WIN32

WINWINDOW

WINFILESYSTEM

RUNTIME

ENGINE

MAINLOOP

RESOURCES

IFILEMOUNT (INTERFACE)

LOOSEFILEMOUNT

ARCHIVEFILEMOUNT

VIRTUALFILESYSTEM

RESOURCEMANAGER

PLANNEDMODULES

RESOURCES/TYPED

IRESOURCELOADER

RESOURCELOADERREGISTRY

JOBS

JOBSYSTEM

INPUT

INPUTSYSTEM

ACTIONMAP

RENDER

RHI

RENDERER

FRAMEGRAPH

SCENE

WORLD

TOOLS

EDITOR

SERIALIZATION

SAVEGAMESYSTEM SERIALIZER

PHYSICS

PHYSICSSYSTEM

AUDIO

ANIMATION

PLATFORM

App provides policies Engine provides mechanisms

NocturneHost (host exe)

Game (future exe)

Editor (future exe)

LogLevel (enum) BuildConfig (header-only)

MallocAllocator SubsystemDesc

mutex sinks Init()

Shutdown() Write()

SetAssertFailHandler() DefaultAssertFailHandler()

deltaSeconds_ Init() secondsSinceStart_

Shutdown() EndFrame() BeginFrame()

DeltaSeconds() SecondsSinceStart()

NowTicks() Frequency()

TicksToSeconds()

Allocate() Deallocate()

totalAllocated_ outstanding_ allocCount_

Deallocate() Allocate() TotalAllocatedBytes()

OutstandingBytes() AllocationCount()

capacity_ offset_ base_

Allocate() Reset() Init()

Capacity() Used()

Register() StartupAll()

ShutdownAll() BuildStartupOrder()

WinPlatform WinFile WinWindowDesc

FileOpenMode FileStat

Create() Destroy() shouldQuit_ hwnd_

ShouldQuit() Handle() WndProc() RequestQuit()

ApplyClientSize() RegisterClassOnce()

ReadFile() CloseFile() OpenFile()

SeekFile() GetFileStat() GetExecutableDirectoryUtf8()

JoinPathUtf8()

EngineConfig

baseAlloc_ debugAlloc_ registry_ frameArena_ alloc_ frameArenaMem_

resources_ Config() initialized_ cfg_ vfs_ ConfigMutable()

Shutdown() Init() SetContentRoot() SetArchivePath() IsConfigMutable() SetOverrideRoot()

BeginFrame() EndFrame() Allocator() TickOnce() Tick() Run()

FrameArena() KillMemory() Resources() InitMemory() VFS() Run() PumpMessagesNonBlocking()

VPath

FileHandle

ResourceID

ResourceHandle

ResourceState (enum)

OpenRead() Close()

Read() Size()

physicalRoot_

archivePath_ entries_ indexFile_

BuildIndex() FindEntry() ComputeEntryDataOffset()

ReadAt()

MountArchive() mounts_ OpenRead() MountLooseDirectory()

MountCount() Close() Read() Size()

ReadAllBytes() ReadAllText()

engine_ worker_ idToIndex running_ vfs_

pending completed records mtx cv

Update() IsReady() Shutdown() Init() RequestBinary()

HasFailed() GetBytes() GetSize() GetError() ValidateHandle_()

LoadOne_() LoaderThreadMain_()

ResourceType ResourceHandleT

ResourceLoadContext ResourceLoadResult

CanLoad() Type()

Load()
FindLoader() RegisterLoader()

JobPriority JobHandle

LockFreeQueue

Start() Stop()

Enqueue() Wait()

Shutdown() BeginFrame() Init()

IsDown() MouseDelta() ProcessOSMessages()

Rebind() Bind()

GetActionValue()

Dx12Device Dx12SwapChain Dx12CommandQueue

DescriptorAllocator GPUResourceAllocator Dx12CommandList

Fence

Shutdown() BeginFrame() Init()

CreateBuffer() CreateTexture() EndFramePresent()

CreateShader() CreatePipelineState()

RenderFrame()

AddPass() Compile()

Execute()

SpatialIndex TransformComponent

CreateEntity() DestroyEntity()

Update()

Cooker Packer

DockingUI ContentBrowser

Inspector SceneHierarchy

Save() Load() WriteBinary() ReadBinary()

Step()

AudioSystem

AnimationSystem

Win32

Main Thread

Resource Loader Thread

Render Thread

Job Worker Threads

Audio Thread

Physics Thread

Windows OS No upward dependencies allowed

all

File I/O

Gameplay

ECS/Component

Memory

Present

Register

MountLooseDirectory

MountArchive

RegisterLoader

AddPass

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